

Effects of dominance and vision on unipedal balance tests in futsal players using a triaxial accelerometer

Juan Guerrero-Henriquez, Guillermo Mendez-Rebolledo, Leandro LLancaleo & Martin Vargas

To cite this article: Juan Guerrero-Henriquez, Guillermo Mendez-Rebolledo, Leandro LLancaleo & Martin Vargas (09 Jan 2024): Effects of dominance and vision on unipedal balance tests in futsal players using a triaxial accelerometer, Sports Biomechanics, DOI: 10.1080/14763141.2024.2301987

To link to this article: <https://doi.org/10.1080/14763141.2024.2301987>



Published online: 09 Jan 2024.



Submit your article to this journal [↗](#)



Article views: 26



View related articles [↗](#)



View Crossmark data [↗](#)



Effects of dominance and vision on unipedal balance tests in futsal players using a triaxial accelerometer

Juan Guerrero-Henriquez ^a, Guillermo Mendez-Rebolledo ^{b,c}, Leandro LLancaleo^a and Martin Vargas^a

^aRehabilitation and Human Movement Sciences Department, Faculty of Health Sciences, University of Antofagasta, Antofagasta, Chile; ^bEscuela de Kinesiología, Facultad de Salud, Universidad Santo Tomás, Talca, Chile; ^cMagíster en Ciencias de la Actividad Física y del Deporte Aplicadas al Entrenamiento, Rehabilitación y Reintegro Deportivo, Facultad de Salud, Universidad Santo Tomás, Talca, Chile

ABSTRACT

Optimal postural control improves performance and reduces the risk of injury in futsal. In this context, wearable accelerometers may detect velocity changes of the centre of mass during a task, enabling the analysis of postural control in different environments. This work aimed to determine the influence of vision and dominance on unipedal static postural balance in non-professional athletes. Twenty-four university male futsal players performed a unipedal balance test to assess their body sway using a triaxial accelerometer. To assess dominance, the preferred limb for kicking the ball was considered, while vision was manipulated by asking participants to close their eyes during the test. Root mean square (RMS) and sample entropy (SaEn) of centre of mass variables were analysed. For statistical analysis, a multivariate analysis of variance model was used. Our results suggest an effect of vision, but not of dominance nor the interaction between vision and limb dominance. Specifically, a higher-acceleration RMS in the mediolateral axis was observed, as well as an increased SaEn in the three axes. To conclude, unipedal postural demand in futsal players under visual input suppression was not influenced by their limb dominance

ARTICLE HISTORY

Received 16 May 2022

Accepted 30 December 2023

KEYWORDS

Postural control; athlete; sport performance

Introduction

The complexity of different actions during the futsal game requires interlimb functional and neuromuscular symmetry as a relevant aspect of player performance and injury prevention (López-Fernández et al., 2020). Specifically, deficits in intrinsic modifiable neuromuscular parameters, such as postural stability during single-leg stance, may potentially increase the risk of non-contact lower extremity injuries in different athletic populations (Dingenen et al., 2016), and neuromuscular training in one-legged weight-bearing conditions causes an improvement in balance and active joint position sense in futsal players (Lopes et al., 2019). Concerning leg dominance, a recent systematic review by DeLang et al. (2021) reported a 1.6 times greater risk of injury on the dominant limb in

soccer players and recommended using consistent methods for assessing limb dominance (DeLang et al., 2021).

One-leg optimal postural control is decisive for futsal technical actions such as passing or kicking the ball. To increase in-game accuracy, these gestures are usually performed by the dominant limb, while the non-dominant controls body weight (Jadczak et al., 2018; Palucci Vieira et al., 2021; Teixeira et al., 2011). To accurately perform these technical actions, the postural strategies to regulate the body's centre of mass (COM) require sensory feedback from its surroundings, acquired through sensory, visual and vestibular systems (Lamoth & van Heuvelen, 2012). In this context, soccer players depend on visual information from the ball and other players' movements (Hammami et al., 2014), while regarding postural strategies and athletic level, professional players are less dependent on vision input when controlling their postural balance (Ben Moussa et al., 2012; Jadczak et al., 2018), prioritising proprioceptive and vestibular inputs. This limb dominance and visual dependence documented support vision manipulation in unipodal postural control assessments (Dębski et al., 2017; Jadczak et al., 2018; Lopes et al., 2019; Paillard et al., 2006).

Multiple sensors can analyse performance in these unipodal tests, such as force platforms, 3D motion capture systems, inertial sensors and triaxial accelerometers. A review conducted by Paillard and Noé (2020) and a recent meta-analysis conducted by Schorderet et al. (2021) indicated that regardless of the method used to evaluate unipodal postural control, an absence of significance in the differences between the legs is observed (Paillard & Noé, 2020; Schorderet et al., 2021). When considering the devices used to determine postural control in the two aforementioned reviews, most of the articles use COP analysis using force platforms and one used accelerometers to analyse static balance (Jadczak et al., 2018). However, accelerometers present potential advantages for balance assessment in athletes, particularly concerning cost and use in ecological environments. In this context, body-worn accelerometers can be useful for detecting kinematics changes in the centre of mass (COM) over time, with satisfactory outcomes in terms of validity and reliability (Pino-Ortega et al., 2019).

From these types of wearable devices, characteristics can be extracted that allow us to describe the highly variable nature of static postural sway. Entropy allows to determine the regularity of postural sway and quantifies the temporal structure of the signal during a task. Previous work has related sample entropy with the adaptive capacity of the motor control system, where less entropy (i.e., more repeating patterns) characterised persons with lower postural control (Manor et al., 2010; Rhea et al., 2019; Stergiou & Decker, 2011; R. Van Emmerik et al., 2016). In our context, this non-linear variable, in conjunction with linear summary statistical variables such as the average characteristic of body COM velocity changes, allows a more accurate understanding of motor control over time (Terada et al., 2018). That is why, knowledge of the internal organisation of balance in the dominant and non-dominant lower extremities, through changes in body COM velocity, could provide information on performance in the discipline and formulation of individualised training or rehabilitation programmes (López-Fernández et al., 2020).

This work aimed to determine the influence of vision and dominance in university futsal players on unipodal postural balance using a triaxial accelerometer. We hypothesise an effect of both conditions (i.e., vision and dominance) in adaptive postural

behaviour (i.e., increased sample entropy and no significant changes in COM acceleration magnitude) in futsal players under closed eyes conditions.

Materials and methods

Participants

This cross-sectional study was conducted in the Human Movement Analysis Laboratory of the Universidad de Antofagasta. This investigation was designed in accordance with the Helsinki Consensus on biomedical research in humans, following the STROBE statement to report observational studies (von Elm et al., 2008). The research protocol was approved by the Scientific Research Ethics Committee of the Universidad de Antofagasta (File No. 371/2022), and an informed consent form was read and signed by each participant. According to a sample size calculation (more detail in Statistical Analysis section), using a convenience sampling method, we recruited futsal players from the futsal team of the Universidad de Antofagasta Sports Service (Deportes UA). Participants were excluded if they reported injury or pain in both lower extremities, history of ligament injury or surgery, or any musculoskeletal or neurological condition that could limit performance in balance tests.

Procedures

Participants completed a health questionnaire prior to participation. They were asked to attend with clothes and footwear they usually wear for sports practice. The measurements were carried out in a single session, on a Monday following a weekend without training or futsal competition.

All participants performed a warm-up on a cycloergometer for 10 min. A triaxial accelerometer (DTS 3D Accelerometer, single radio module with 3 sub channels, 400 g operational range, Noraxon Inc., Scottsdale, AZ, USA) was placed at the level of the second sacral vertebra by the same researcher as a point representing the body COM. The Y axis was oriented in the vertical axis direction (V), X in the medio-lateral direction (ML) and Z in the antero-posterior direction (AP).

Each participant was asked to stand as still as possible while performing three trials of 30-second unipodal balance test, with their dominant and non-dominant lower extremity, under opened (OE) and closed eyes (CE) conditions. Dominant lower extremity was defined as the one that each participant uses to kick the ball. Participants were instructed to keep their upper extremities crossed against the chest and to raise the lower limb—contralateral to the one evaluated—when a sound signal was emitted. For OE condition, they were asked to look attentively at a mark in front of them, located at 2.5 metres at their eyes level, supervised by an investigator during the execution of each trial. The order of the trials was randomised across participants to avoid learning bias between tests and data analysis. A 30-second rest was given between trials.

Signal acquisition and processing

The acceleration data were recorded telemetrically with a sampling rate of 1,500 Hz (Desktop DTS system, Noraxon Inc., Scottsdale, AZ, USA). Subsequently, raw data was

imported into Matlab R2017b (The MathWorks Inc., Natick, MA, USA) where signal processing was performed.

The first and last 5 seconds of the recording were removed (Neville et al., 2015). A central trend line removal was applied to the centre around zero. Finally, a fourth-order Butterworth low-pass filter with a cut-off frequency of 10 Hz was applied.

Measures

From the acceleration signals per axis (i.e., ML, AP and V), the root mean square (RMS) and sample entropy (SaEn) were extracted as linear and nonlinear variables, respectively. RMS represents the mean variance of a captured signal during a period of time (Pino-Ortega et al., 2019) and has been widely used for the description of postural performance in sports science (Johnston et al., 2019).

SaEn provides insight on the regularity of the temporal structure of body COM acceleration time series (Richman & Moorman, 2000). More irregular signals have been associated with better health and performance (Vaz et al., 2020; Walsh et al., 2018). As described by Richman and Moorman (2000), the acceleration series are first embedded in an m -dimensional delayed space, where the vectors are constructed as:

$$x(p) = [x(p+k)]_{k=0}^{m-1}, p = 1, 2, \dots, n-m+1$$

The probability $B^m(r)$ that two sequences match for m points is computed by counting the average number of vector pairs, for which the distance is lower than the tolerance r . Similarly, $A^m(r)$ is defined for an embedding dimension of $m+1$. The SaEn is then calculated as:

$$SaEn(x, m, r) = -\ln(A^m(r)/B^m(r))$$

For SaEn calculation, three parameters were defined: the amount of data in the acceleration signal (n); the length of the sequences to be compared (m); and a tolerance parameter for accepting the matches (r). For the purposes of this study, $n = 45,000$ samples per signal; $m = 2$; and $r = 0.2$ were defined, given that the deviation error is less than 3% for signals with $n > 100$ samples and which also have a mean tending towards 0 (Richman & Moorman, 2000).

Statistical analysis

Sample size calculation was performed in G*Power version 3.1.9.4. (G*Power, Version 3.0.10, Franz Faul, Universität Kiel, Kiel, Germany) (Faul et al., 2007), following previous recommendations (Grissom & Kim, 2014). Considering a Pillai-Bartlett trace value of 0.4, a significance level $\alpha = 0.05$ and a power $\beta = 0.80$, a sample of 24 subjects was required to find associations between numerical variables. Further statistical analysis was performed using SPSS v.25 (IBM Corp., Armonk, New York).

Sample characteristics were presented as mean and standard deviation for continuous variables and as simple frequency and percentage for categorical variables. Shapiro-Wilk and Levene's test were applied to determine data distribution and homogeneity of variances, respectively. To determine the effects of dominance (2: D and ND) and visual

manipulation (2: EO and CE) effect on COM acceleration outcomes (RMS and SaEn by direction), a multivariate analysis of variance was applied. The follow-up of was carried out by means of univariate contrasts. The eta squared statistic (η^2) was reported as a measure of the effect size. Finally, in the case of significant effects, pairwise comparisons were carried out with Bonferroni correction. For all statistical tests, an alpha of 0.05 was used.

Results

Sample characteristics

Twenty-four male futsal players were recruited. The age was 23.51 ± 2.21 years, weight = 73.44 ± 8.02 kg, height = 1.73 ± 0.06 m; BMI = 24.53 ± 1.73 kg/m². Average futsal practicing experience was 1.42 ± 0.50 years and 6 (25%) players from the sample played in defence position, 9 (37.50%) were forwards, 7 (29.17%) were midfielders and 2 (8.33%) were goalkeepers.

Multivariate analysis of variance

A vision effect was found ($F_{11;36} = 4.608$; $p < 0.001$; $\eta^2 = 0.997$) on the dependent variables. No significant effects of dominance were found ($F_{11;36} = 1.247$; $p = 0.293$; $\eta^2 = 0.558$) nor of the interaction vision $\times$ dominance ($F_{11;36} = 1.365$; $p = 0.231$; $\eta^2 = 0.606$).

Concretely, the effect of the visual manipulation was observed on the RMS ML ($F_{1;46} = 10.834$; $p = 0.002$; $\eta^2 = 0.897$), SaEn ML ($F_{1;46} = 35.691$; $p < 0.001$; $\eta^2 = 1.000$), SaEn AP ($F_{1;36} = 4.500$; $p = 0.039$; $\eta^2 = 0.547$) and SaEn V ($F_{1;36} = 18.941$; $p < 0.001$; $\eta^2 = 0.989$). Pairwise comparisons are shown in [Figures 1 and 2](#).

Specifically, significant differences were evidenced when manipulating the visual input in the RMS of ML axis in the dominant (EO: 0.1206 ± 0.05 G vs CE: 0.1635 ± 0.09 G; $p = 0.011$) and non-dominant limb (EO: 0.2901 ± 0.07 G vs CE: 0.3213 ± 0.07 G; $p = 0.045$). Moreover, in relation to SaEn, differences were evidenced between eyes open and eyes closed conditions in the AP axis (EO: 0.0229 ± 0.01 a.u. vs CE: 0.0241 ± 0.01 a.u.; $p = 0.002$), ML (EO: 0.0132 ± 0.05 a.u. vs CE: 0.0205 ± 0.01 a.u.; $p < 0.001$) and V (EO: 0.0195 ± 0.01 a.u. vs CE: 0.0231 ± 0.01 a.u. $p < 0.001$) in the dominant limb, while in the non-dominant limb, differences were found in the AP (OE: 0.0229 ± 0.01 a.u. vs CE: 0.0249 ± 0.01 a.u.; $p = 0.010$) and ML (EO: 0.0142 ± 0.01 a.u. vs CE: 0.0196 ± 0.01 a.u.; $p < 0.001$) axes.

Discussion and implications

This research aimed to determine the influence of vision and dominance in university futsal players on unipodal postural balance tests using a triaxial accelerometer. Our results suggest that dominance does not affect the variables analysed, being only possible to evidence an effect of visual manipulation on the RMS and SaEn of the body COM acceleration signals ([Figures 1 and 2](#)).

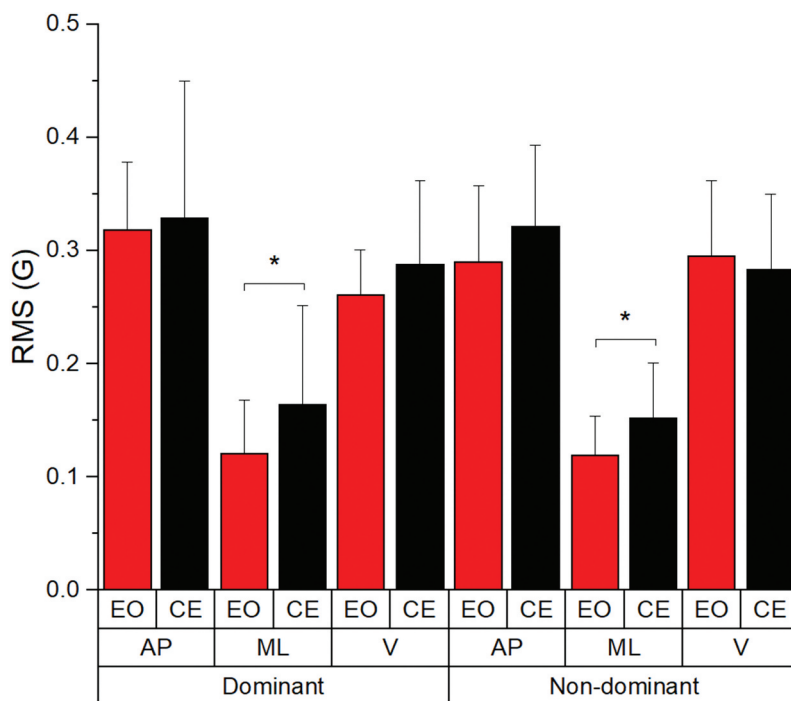


Figure 1. Pairwise comparison for root mean square (RMS) values for dominant and non-dominant lower extremities. ML: medio-lateral; AP: anterior-posterior; V: Vertical; EO: eyes open; CE: closed eyes. *Indicates statistical significance differences.

The visual input manipulation challenged the futsal player's postural control system, reflected by increased body COM acceleration variability (i.e., higher ML RMS) and decreased regularity during the EC task (i.e., higher SaEn magnitude) (Vaz et al., 2020). The predominance of this response in the ML axis may arise because multi-joint adjustments on narrow surfaces are predominantly performed relative to the hip, by trunk acceleration and upper limb movements in the ML direction (Van Dieën et al., 2015). Specifically, the increased variability in ML COM acceleration can be explained by the higher level of hip and knee control required by the task with EC, expressed through greater recruitment of distal muscle synergies observed in healthy subjects with optimal postural performance in single-leg tasks (Ghislieri et al., 2020). Previous research has related entropy with the adaptive capacity of the motor control system (Manor et al., 2010; Stergiou & Decker, 2011; R. E. A. van Emmerik et al., 2016). In this context, body sway tends to be less regular during EC conditions as a manifestation of motor reorganisation (Borg & Laxaback, 2010), or when an external attentional focus is adopted (Rhea et al., 2019). Moreover, the less regular and more variable body sway acceleration in one of the three directions can be interpreted as a good performance close to an optimal postural adjustment (Jadczak et al., 2018; Lamothe & van Heuvelen, 2012), reflecting an adaptable and flexible postural control (Kędziołek & Błażkiewicz, 2020). Although our results in SaEn showed sensitivity to visual feedback, participants' attentional focus during the task must be addressed. In our protocol, the futsal player's focus of attention

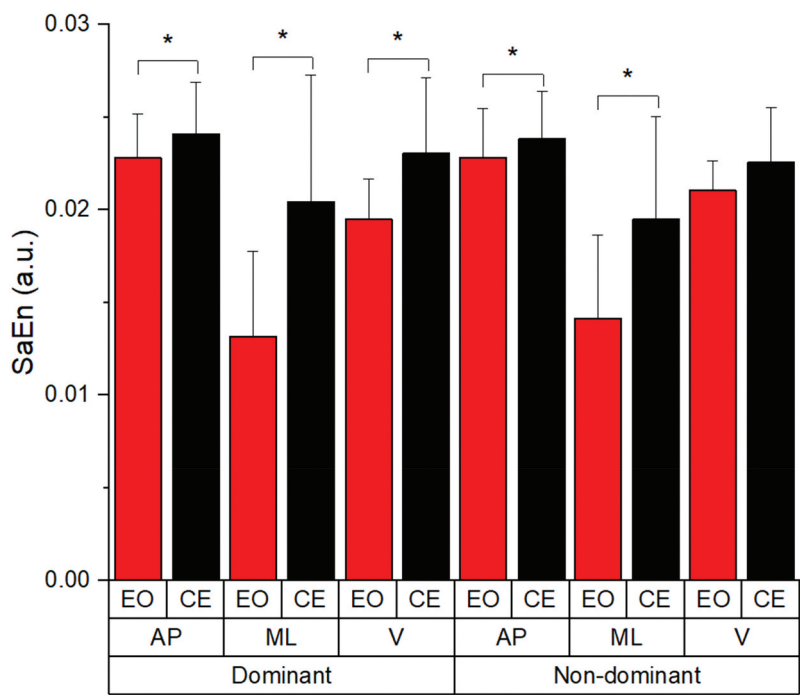


Figure 2. Pairwise comparison for SaEn (sample entropy) values for dominant and non-dominant lower extremities. a.u.: adimensional units; ML: medio-lateral; AP: anterior-posterior; V: Vertical; EO: eyes open; CE: closed eyes. *Indicates statistical significance differences.

was internal under visual manipulation (i.e., while keeping a unipodal posture), which may have mobilised cognitive involvement and influenced the postural control response (Donker et al., 2007). A final aspect to consider, regarding vision, is that although it has been shown that futsal players with more years of practice show less visual dependence in their postural balance, they are less dependent on their eyes (Ben Moussa et al., 2012; Jadczyk et al., 2018). The extent of our results does not allow us to discuss this matter; however, future research could compare this population with non-practicing subjects, in order to show if there are differences in postural dependence in vision.

Concerning the absence of dominance observed, our results are similar according to King and Wang (2017) and to Paillard et al. (2006) and can be related to the specificity of the sport which requires fine anteroposterior postural control (King & Wang, 2017; Paillard et al., 2006). Other possible explanations for the absence of a dominance effect could be the controlled characteristics of the laboratory measurements, in which game-specific factors could be attenuated (Dębski et al., 2017). In this context, a review conducted by Schorderet et al. (2021) showed no differences between dominant and non-dominant lower extremities in static balance tests (Schorderet et al., 2021).

Within the implications of this research, the college futsal players exhibited postural control visual dependency but no limb dominance in unipodal static balance tests. As a hypothesis, under visual dependency, an expected postural control strategy in unipodal balance could be characterised by increased mean-variance magnitude changes only in

ML, and less overall regularity in COM acceleration time series. Future research could be based on this approach to assess if unipodal postural behaviour is a potential non-contact lower limb injury risk factor and to further analyse the effect of postural stability training on these variables in different populations.

We must state some limitations in the present investigation. The definition of the dominant limb is not straightforward in futsal because there is a constant exchange of functions between limbs. Extrapolation of results to other populations should be done with care and considering disciplines of similar performance. For signal processing, acceleration series of shorter duration (i.e., 20 seconds) were considered; however, a systematic review with meta-analysis by Schorderet et al. (2021) highlights research on unipodal balance tests with 10- to 20-second analysis (Schorderet et al., 2021). Finally, in laboratory postural test conditions, attention is placed on the motor task to maintain body balance as best as possible; therefore, we recommend performing balance assessment tasks that involve participant's attention in similar conditions to the futsal game.

Conclusion

We can conclude that, under unipodal postural tasks with visual manipulation, non-injured, college futsal players adjust their posture, increasing mediolateral acceleration variability and decreasing overall regularity in body COM acceleration. Furthermore, no limb dominance effect was found.

Acknowledgments

The authors would also like to thank José Casareggio, technical director of the futsal team, and Daniel Orrego, Director of Deportes UA, for facilitating access to the participants.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Juan Guerrero-Henriquez  <http://orcid.org/0000-0002-9071-3586>
Guillermo Mendez-Rebolledo  <http://orcid.org/0000-0001-7948-3807>

References

- Ben Moussa, A. Z., Zouita, S., Dziri, C., & Salah, F. Z. B. (2012). Postural control in Tunisian soccer players. *Science et Sports*, 27(1), 54–56. <https://doi.org/10.1016/j.scispo.2011.03.006>
- Borg, F., & Laxaback, G. (2010). Entropy of balance – Some recent results. *Journal of NeuroEngineering and Rehabilitation*, 7(38). <https://doi.org/10.1186/1743-0003-7-38>
- Dębski, P., Mielńczyk, J., & Gnat, R. (2017). Postural stability in one leg standing on dominant and non-dominant leg in professional soccer players. *Journal of Kinesiology and Exercise Sciences*, 26(77), 53–64. <https://doi.org/10.5604/01.3001.0010.7105>

- DeLang, M., Salamh, P., Farooq, A., Tabben, M., Whiteley, R., van Dyk, N., & Chamari, K. (2021). The dominant leg is more likely to get injured in soccer players: Systematic review and meta-analysis. *Biology of Sport*, 38(3), 397–435. <https://doi.org/10.5114/biolsport.2021.100265>
- Dingenen, B., Malfait, B., Nijs, S., Peers, K., Vereecken, S., Verschueren, S., Janssens, L., & Staes, F. (2016). Postural stability during single-leg stance: A preliminary evaluation of noncontact lower extremity injury risk. *Journal of Orthopaedic & Sports Physical Therapy*, 46(8), 650–657. <https://doi.org/10.2519/jospt.2016.6278>
- Donker, S., Roerdink, M., Greven, A., & Beek, P. (2007). Regularity of center-of-pressure trajectories depends on the amount of attention invested in postural control. *Experimental Brain Research*, 181(1), 1–11. <https://doi.org/10.1007/s00221-007-0905-4>
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G*power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/BF03193146>
- Ghislieri, M., Knaflitz, M., Labanca, L., Barone, G., Bragonzoni, L., Benedetti, M., & Agostini, V. (2020). Muscle synergy assessment during single-leg stance. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(12), 2914–2922. <https://doi.org/10.1109/TNSRE.2020.3030847>
- Grissom, R. J., & Kim, J. J. (2014). *Effect sizes for research: Univariate and multivariate applications*. (1st ed.). Routledge. <https://doi.org/10.4324/9781410612915>
- Hammami, R., Behm, D. G., Chtara, M., Othman, A. B., & Chaouachi, A. (2014). Comparison of static balance and the role of vision in elite athletes. *Journal of Human Kinetics*, 40(1), 33–41. <https://doi.org/10.2478/hukin-2014-0030>
- Jadczak, Ł., Grygorowicz, M., Dzudziński, W., & Śliwowski, R. (2018). Comparison of static and dynamic balance at different levels of sport competition in professional and junior elite soccer players. *Journal of Strength and Conditioning Research*, 33(12), 3384–3391. <https://doi.org/10.1519/JSC.0000000000002476>
- Johnston, W., Reilly, M. O., Argent, R., & Caulfield, B. (2019). Reliability, validity and utility of inertial sensor systems for postural control assessment in sport science and medicine applications : A systematic review. In *Sports medicine*. Springer International Publishing. <https://doi.org/10.1007/s40279-019-01095-9>
- Kędziorek, J., & Błażkiewicz, M. (2020). Nonlinear measures to evaluate upright postural stability: A systematic review. *Entropy*, 22(12), 1–24. <https://doi.org/10.3390/e22121357>
- King, A., & Wang, Z. (2017). Asymmetrical stabilization and mobilization exploited during static single leg stance and goal directed kicking. *Human Movement Science*, 54, 182–190. <https://doi.org/10.1016/j.humov.2017.05.004>
- Lamoth, C. J. C., & van Heuvelen, M. J. G. (2012). Sports activities are reflected in the local stability and regularity of body sway: Older ice-skaters have better postural control than inactive elderly. *Gait and Posture*, 35(3), 489–493. <https://doi.org/10.1016/j.gaitpost.2011.11.014>
- Lopes, M., Lopes, S., Patinha, T., Araújo, F., Rodrigues, M., Costa, R., Oliveira, J., & Ribeiro, F. (2019). Balance and proprioception responses to FIFA 11+ in amateur futsal players: Short and long-term effects. *Journal of Sports Sciences*, 37(20), 2300–2308. <https://doi.org/10.1080/02640414.2019.1628626>
- López-Fernández, J., García-Unanue, J., Sánchez-Sánchez, J., Colino, E., Hernando, E., & Gallardo, L. (2020). Bilateral asymmetries assessment in elite and sub-elite male futsal players. *International Journal of Environmental Research and Public Health*, 17(9), 3169. <https://doi.org/10.3390/ijerph17093169>
- Manor, B., Costa, M., Hu, K., Newton, E., Starobinets, O., Kang, H., Peng, C., Novak, V., & Lipsitz, L. (2010). Physiological complexity and system adaptability: Evidence from postural control dynamics of older adults. *Journal of Applied Physiology*, 109(6), 1786–1791. <https://doi.org/10.1152/japplphysiol.00390.2010>
- Neville, C., Ludlow, C., & Rieger, B. (2015). Measuring postural stability with an inertial sensor: Validity and sensitivity. *Medical Devices: Evidence and Research*, 8, 447–455. <https://doi.org/10.2147/MDER.S91719>

- Paillard, T., & Noé, F. (2020). Does monopodal postural balance differ between the dominant leg and the non-dominant leg? A review. *Human Movement Science*, 74(June), 102686. <https://doi.org/10.1016/j.humov.2020.102686>
- Paillard, T., Noé, F., Riviére, T., Marion, V., Montoya, R., & Dupui, P. (2006). Postural performance and strategy in the unipedal stance of soccer players at different levels of competition. *Journal of Athletic Training*, 41(2), 172–176.
- Palucci Vieira, L. H., Kalva-Filho, C. A., Santinelli, F. B., & Clemente, F. M. (2021). Lateral preference and inter-limb asymmetry in completing technical tasks during official professional futsal matches: The role of playing position and opponent quality. *Frontiers in Psychology*, 12 (August), 1–11. <https://doi.org/10.3389/fpsyg.2021.725097>
- Pino-Ortega, J., Hernández-Belmonte, A., Gómez-Carmona, C., Bastida-Castillo, A., García-Rubio, J., & Ibáñez, S. (2019). Monopodal postural stability assessment by wireless inertial measurement units through the fast Fourier transform. *Journal of Sport Rehabilitation*, 29(6), 738–747. <https://doi.org/10.1123/jsr.2018-0399>
- Rhea, C. K., Diekfuss, J. A., Fairbrother, J. T., & Raisbeck, L. D. (2019). Postural control entropy is increased when adopting an external focus of attention. *Motor Control*, 23(2), 230–242. <https://doi.org/10.1123/mc.2017-0089>
- Richman, J., & Moorman, R. (2000). Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology Heart and Circulatory Physiology*, 278(6), H2039–H2049. <https://doi.org/10.1152/ajpheart.2000.278.6.H2039>
- Schorderet, C., Hilfiker, R., & Allet, L. (2021). The role of the dominant leg while assessing balance performance. A systematic review and meta-analysis. *Gait and Posture*, 84, 66–78. <https://doi.org/10.1016/j.gaitpost.2020.11.008>
- Stergiou, N., & Decker, L. M. (2011). Human movement variability, nonlinear dynamics, and pathology: Is there a connection? *Human Movement Science*, 30(5), 869–888. <https://doi.org/10.1016/j.humov.2011.06.002>
- Teixeira, L. A., de Oliveira, D. L., Romano, R. G., & Correa, S. C. (2011). Leg preference and interlateral asymmetry of balance stability in soccer players. *Research Quarterly for Exercise and Sport*, 82(1), 21–27. <https://doi.org/10.1080/02701367.2011.10599718>
- Terada, M., Kosik, K., Johnson, N., & Gribble, P. (2018). Altered postural control variability in older-aged individuals with a history of lateral ankle sprain. *Gait & Posture*, 60(October), 88–92. <https://doi.org/10.1016/j.gaitpost.2017.11.009>
- Van Dieën, J. H., Leeuwen, M., & Faber, G. (2015). Learning to balance on one leg: Motor strategy and sensory weighting. *Journal of Neurophysiology*, 114(5), 2967–2982. <https://doi.org/10.1152/jn.00434.2015>
- van Emmerik, R. E. A., Ducharme, S. W., Amado, A. C., & Hamill, J. (2016). Comparing dynamical systems concepts and techniques for biomechanical analysis. *Journal of Sport and Health Science*, 5(1), 3–13. <https://doi.org/10.1016/j.jshs.2016.01.013>
- Van Emmerik, R., Ducharme, S. W., Amado, A. C., & Hamill, J. (2016). Comparing dynamical systems concepts and techniques for biomechanical analysis. *Journal of Sport and Health Science*, 5(1), 3–13. <https://doi.org/10.1016/j.jshs.2016.01.013>
- Vaz, J. R., Stergiou, N., Diniz, A., Dinis, R., & Pezarat-Correia, P. (2020). Postural control is altered in females with excessive medial knee displacement. *Sports Biomechanics*, 22(7), 848–862. <https://doi.org/10.1080/14763141.2020.1767187>
- von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Gøtzsche, P. C., & Vandenbroucke, J. P. (2008). The strengthening the reporting of observational studies in epidemiology (STROBE) statement: Guidelines for reporting observational studies. *Journal of Clinical Epidemiology*, 61 (4), 344–349. <https://doi.org/10.1016/j.jclinepi.2007.11.008>
- Walsh, M., Slattery, E., McMath, A., Cox, R., & Haworth, J. (2018). Training history constrains postural sway dynamics: A study of balance in collegiate ice hockey players. *Gait and Posture*, 66 (September), 278–282. <https://doi.org/10.1016/j.gaitpost.2018.09.009>